Motivation to Move from ANN/MLP to Graph Neural Networks (GNNs)

1. Limitation of ANN/MLP

Artificial Neural Networks (ANNs), particularly Multi-Layer Perceptrons (MLPs), work well when data is structured and tabular, where every example is an independent, fixed-length feature vector. The Cora dataset, however, is not just a table of paper features — it is a citation network, where papers are connected to one another through citation links.

2. Why Relational Data Matters

This relational structure is exactly what an MLP cannot use. An MLP would classify each paper using only its 1433-dimensional bag-of-words vector, completely ignoring which papers it cites or is cited by. In domains such as citation networks, social networks, and molecular structures, entities are connected through meaningful relationships, and MLPs process each instance independently, discarding this structural information entirely.

3. How GNNs Solve This: Message Passing

Graph Neural Networks represent data as a graph, where nodes are entities (papers) and edges are relationships (citations). A GNN introduces message passing: each node updates its representation by aggregating information from its neighbours. The 2-layer GCN used here computes $H_{out} = A_{hat} \cdot H_{in} \cdot W$, where A_{hat} is the symmetrically normalised adjacency matrix, so a node's new features reflect both its own content and its local citation context.

4. Handling Irregular Graph Structures

Unlike images or tables, graphs have no fixed size and no natural ordering of neighbours — a node may have two citations or twenty, and the order in which they are listed should not change the result. Conventional MLPs assume a fixed input shape and cannot handle this naturally, whereas GNNs are built specifically to operate on this kind of structure.

5. Impact on Cora Results

This difference has a direct, measurable effect: an MLP using only node features typically reaches around 55–60% test accuracy, while the 2-layer GCN reaches roughly 80–82% on the same split. The improvement comes entirely from exploiting the citation graph, not from a larger model.

Conclusion

- MLPs work well on independent, tabular data but cannot capture relationships between examples.
- Cora's citation links carry predictive signal that MLPs completely ignore.
- GNNs overcome this through message passing, aggregating each node's neighbourhood.
- GNNs are built for irregular, variable-sized, unordered graph structures.
- Result: GCN reaches ~80–82% accuracy on Cora vs ~55–60% for an MLP, making GNNs the appropriate choice for node classification on graph data.